

Strong continuity is unnecessary in the local zero–two law

Literature-status note for run `fa_banach_001`

17 August 2026

Status: literature already answered. Chojnacki proves the zero–two law for arbitrary normed-algebra-valued cosine families, removing all continuity assumptions; Esterle independently gives a short Banach-algebra proof.

Source question

Schwenninger and Zwart prove that a strongly continuous cosine family $C : \mathbb{R} \rightarrow \mathcal{L}(X)$ satisfies

$$\limsup_{t \rightarrow 0} \|C(t) - I\| < 2 \implies \lim_{t \rightarrow 0} \|C(t) - I\| = 0.$$

Immediately after Theorem 1.1 on PDF page 2 of arXiv:1402.1304 [1], they write that it remains open whether strong continuity can be removed.

Decisive answer

Chojnacki’s paper explicitly announces that it extends the two Schwenninger–Zwart results to cosine families with no continuity assumption. Theorem 3 on PDF page 8 states [2]:

Let A be a normed algebra with unity e , and let $c = \{c(t)\}_{t \in \mathbb{R}}$ be a cosine family in A such that $\limsup_{t \rightarrow 0} \|c(t) - e\| < 2$. Then $\lim_{t \rightarrow 0} c(t) = e$.

Taking $A = \mathcal{L}(X)$ gives exactly the source conclusion, without strong continuity, measurability, or any other regularity hypothesis.

Esterle’s separate arXiv paper [3] explicitly identifies the same Schwenninger–Zwart question, notes Chojnacki’s affirmative answer, and gives an independent short direct proof. Its Theorem 3.2 on PDF page 6 proves the same implication for a cosine function in a unital Banach algebra.

Status, scope, and search evidence

This is a complete affirmative literature answer, not a new result of the run. Chojnacki proves a stronger normed-algebra theorem, while Esterle gives an independent Banach-algebra proof. Exact arXiv-id, title, quoted-question, strong-continuity-removal, and general-cosine-function searches found these explicit matches. The local run indexes contained no prior packet for this question; the existing scaled/global cosine-family packet addresses a different problem.

References

- [1] F. L. Schwenninger and H. Zwart, *Zero-two law for cosine families*, J. Evol. Equ. 15 (2015), 559–569; arXiv:1402.1304.
- [2] W. Chojnacki, *Around Schwenninger and Zwart’s zero-two law for cosine families*, Arch. Math. (Basel) 106 (2016), 561–571; doi:10.1007/s00013-016-0898-y.
- [3] J. Esterle, *A short proof of the zero-two law for cosine functions*, arXiv:1505.06065 (2015).